# Merzbacher Comparison — recipe and their true baseline

## 2026.07.27 — Read this before quoting any comparison number

Note crumbs so the head-to-head can be run cold. The protocol is fixed **before** we have a model, so it is not chosen after seeing our number.

| what | where |
|---|---|
| split builder | `experiments/020-cachera-betaxanthin/scripts/build_merzbach` |
| split output | `results/merzbacher_nested_split.json` |
| baseline analysis | `experiments/020-cachera-betaxanthin/scripts/analyze_merzba` |
| baseline output | `results/merzbacher_baseline_analysis.json` |
| their archive (sha256-pinned) | `$DATA_ROOT/data/merzbacher2025_fcl/` |

### What their deposit gives us — far more than the paper implies

The manuscript reports no split and an inconsistent test N (Fig 4b 659, Table S6 649, 20 % of 811 = 162). The Zenodo **code** deposit (10.5281/zenodo.15518666 → record 15761895) has:

| file | what it gives us |
|---|---|
| `data/yeast_production_test_split.csv` | the exact 640-ORF test set |
| `data/yeast_production_validation_split.csv` | same genes + released `label` 0/1/2 + CV `fold` |
| `training/yeast_training_production.py` | the exact binning rule |
| `figures/fig4/fig4b.csv` | per-fold accuracy, macro-F1, **MCC**, class-2 accuracy |
| `figures/fig4/fig4c_*.csv` | **per-gene, per-flux-sample predictions + class scores** |

Only the 49 MB code archive is mirrored; `data.zip` is 23 GB of flux samples we do not need.

### THEIR TRUE BASELINE — from their own shipped predictions

Aggregated to gene level by majority vote over each deletion's flux samples, exactly as their `tools/knockout_voting.py` does. Best model, `RandomForestClassifier_Resampled`:

| true ↓ / pred → | low | **medium** | high |
|---|---|---|---|
| low (109) | 6 | **101** | 2 |
| medium (431) | 2 | **424** | 5 |
| high (100) | 0 | **82** | 18 |

- gene-level accuracy **0.700**, majority-class rate **0.673**
- **607 of 640 genes called medium — 94.8 %**
- **18 of 100 high producers found**
- majority predictor gets 431/640; theirs gets 448/640. **The entire margin is 17 genes.**

**They computed MCC and did not report it.** From `fig4b.csv`:

| model | accuracy | MCC | high-producer acc |
|---|---|---|---|
| RandomForest | 0.698 | **0.232** | 0.181 |
| RandomForest_Balanced | 0.698 | 0.233 | 0.117 |
| LogisticRegression | 0.515 | **0.041** | 0.233 |
| LinearSVC_Resampled | 0.445 | **0.031** | 0.273 |

MCC 0.23 is weak but **not zero** — there is real signal, concentrated in the majority class. Be fair in any writeup:

they claim “promising accuracy”, not a significant gain.

**The decisive structural point:** every model that finds more high producers has *worse* overall accuracy, because the only way to call more highs is to stop calling everything medium. **Accuracy actively selects against the capability the task is about.**

#### The recipe, fixed in advance

1. **Ground truth = THEIR RELEASED LABELS**, never labels we re-derive. Their rule min-max scales to [0,1] then cuts at 0.40 / 0.65, so the cuts depend on the observed extremes. Applied to our (larger) copy of the same screen it gives 107/476/56 against their 109/431/100 — 81.2 % agreement, and **we call barely half as many high producers**. Same rule, same screen, different classes. Re-deriving would compare different tasks.
2. **Bin OUR predictions with thresholds fitted on the TRAIN split only.** Fitting on test leaks the class distribution, which on a 67 %-majority problem is most of the answer.
3. **Primary metrics are imbalance-immune**: Spearman, and top-*k* enrichment on high producers. Report **MCC** next to accuracy — they had it, so we should show it.
4. **Also report accuracy + per-class + high-producer recall**, so the comparison lands on their terms too.
5. **State the fraction of genes we call medium.** If ours is also ~95 %, we reproduced their failure mode rather than beat it, and must say so.

#### Reconciliations to carry into any writeup — report, never resolve silently

- **639 of 640** of their test genes are in our test split. The one missing is `YBR011C` = **IPP1**, which is **essential** (SGD S000000215, null inviable). No haploid deletion collection contains *ipp1*Δ, so no screen could have measured it — **their test set contains a gene with no possible deletion phenotype.**
- **`PPA1` is an alias of both IPP1 and VMA16.** The screen has ONE `PPA1` row; their split has BOTH YBR011C and YHR026W as separate test genes. At least one of their test entries plausibly carries **VMA16’s** value scored as if it were IPP1. Our screen-local disambiguation (`PPA1`→VMA16, `FEN1`→ELO2) is documented in the split builder, with the reasoning recorded so it can be re-derived if its premises change.
- **640 vs the paper’s 811** — a 171-gene gap the Methods do not count (“sampling failed to converge”).
- **Their validation folds appear drawn from the same 640 genes as their test list** (`yeast_training_production.py:` which would mean model selection touched test data. FLAGGED, not claimed — confirming needs `yeast_single_knockouts.npz` from the 23 GB archive.
- Our Cachera build is **stale** w.r.t. the name resolver (issue #195). The split is built from the RAW screen + resolver so it does not inherit that, but **training data still will** until a rebuild.

#### What we can report that they cannot

- **Regression at all.** They tried and abandoned it — *“challenging with the limited number of knockouts at the high and low ends.”* Our betaxanthin noise ceiling is **r = 0.914** (reliability 0.836, median 15 replicates), so regression is available to us.
- **The ~3,900 non-metabolic deletions.** FCL has no flux cone for a gene outside Yeast9, so it cannot make a prediction at all. We predict all ~4,700. That is a category difference, not a tie-break.

#### Where our model stood when the sweeps were paused

`020-cachera-betaxanthin`, study `betaxanthin_002`, 33/40 trials before cancellation: **val Pearson 0.430** — 47 % of the 0.914 ceiling — with `prot_T5_all` / learnable **False** / `crps` / L=2 / hidden 90 / 6.5e-5 / decayed / yeo-johnson. The top-5 configs agreed closely, so it is a plateau rather than a lucky trial. Companion arms: `021-ozaydin-beta-carotene` (Spearman 0.223) and `022-mulleder-metabolome` (0.180, up from a historical ≤ 0.025).

Related: [[plan.cgt-metabolism-flux-layer.2026.07.26]] · [[plan.simb-2026-multimodal-cgt.2026.07.21]]

## 2026.08.02 - CGT vs Flux Cone Learning on the Cachera betaxanthin screen

**Naming, because the two get conflated.** *Cachera* is the genome-wide betaxanthin DELETION SCREEN – the data, ~4,700 single deletions, and the ground truth for both models. *Flux Cone Learning* (FCL) is Merzbacher et al.’s METHOD, which learns from flux samples drawn through a genome-scale metabolic model;

`RandomForest_Resampled` is its best variant and is what "FCL RF" means throughout. The comparison is **CGT vs FCL, both scored on the Cachera screen** – not "us vs Cachera".

All figures regenerate from `experiments/020-cachera-betaxanthin/scripts/plot_merzbacher_comparison.py`, built from the 10 finished Delta `020_v4` cells (`$DATA_ROOT/test-predictions/`) and the FCL paper's shipped `figures/fig4/fig4c_*.csv`.

- **639 genes** carry their label, a CGT prediction and a raw screen value. Their 640th (IPP1/YBR011C) is essential – no deletion strain exists, so no screen could measure it.
- **FCL model = RandomForest_Resampled**, the best of their four (the others read 0.56-0.64 gene-level against RF's 0.700).
- **CGT = ONE cell, selected by VALIDATION** (`s09_L6_maskon_lr0.0001_yj`, val 0.3639), never by its score on their test genes.
- **Provenance asserted:** our re-derivation of their deployed gene-level vote reproduces their published accuracy (0.7011 vs 0.700) or the script exits.

### TL;DR

1. **The published accuracy comparison is empty.** 0.701 vs a 0.674 majority rate, achieved by calling 95 % of genes medium. Rank-matched, both models fall to ~0.55. (Fig 2)
2. **Both models DO have real signal, concentrated at the top of the ranking.** ~5x enrichment in the top 10-25 genes, $p < 1e\text{-}06$ – and essentially nothing across the bulk. (Fig 3, 4)
3. **CGT and their RF are roughly tied where both can be measured**, and disagree about which genes (rank correlation -0.128). (Fig 1, 3, 5)
4. **The real difference is COVERAGE, not accuracy.** yeast-GEM covers only **915 of the 4,721** deletions in the screen (19.4 %), and **163 of the 223 high producers (73 %) lie outside it** – FCL has no features for any of them. CGT scores those genes and finds high producers there at 4.4x enrichment ($p = 2e\text{-}05$). (Fig 8, 9)

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### Fig 1 – per-gene spread (639 points)

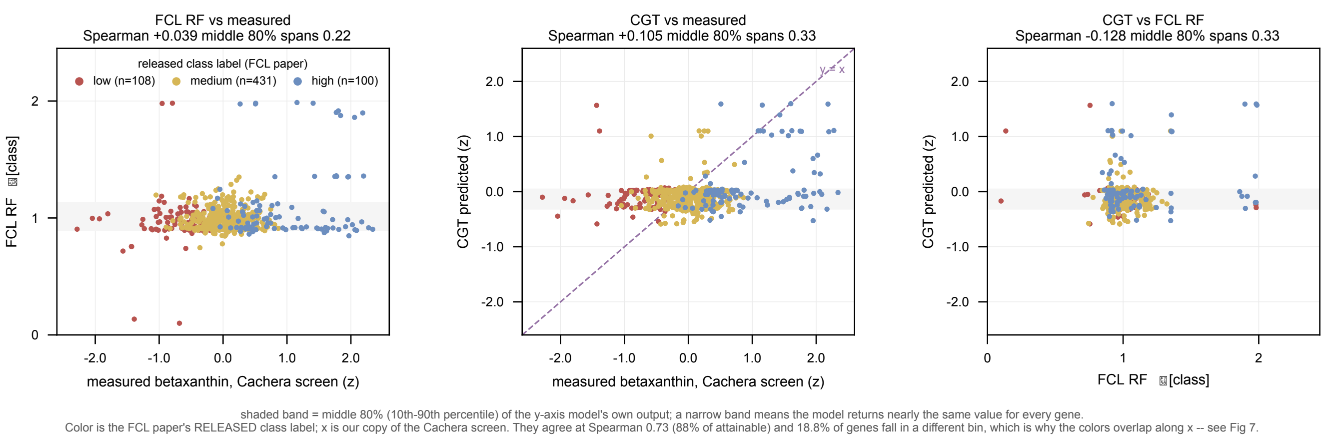


**Fig 1. Both models emit a nearly constant value; the two disagree about which genes.** 80 % of FCL's genes fall inside an $\mathbb{E}$[class] band of 0.22 on a 0-2 scale, 80 % of CGT's inside 0.33 z-units. CGT's prediction is drawn on the MEASUREMENT's own scale (both are z-scored betaxanthin), so the middle panel shows the compression directly: the cloud is a flat band far below the `y = x` perfect-prediction line rather than tracking it. Colour is the FCL paper's released class label, x is OUR copy of the same screen.

| pair | Pearson | Spearman |
|---|---|---|
| measured vs FCL RF | +0.244 | **+0.039** |
| measured vs CGT | +0.294 | **+0.105** |
| FCL RF vs CGT | +0.108 | **-0.128** |

Pearson exceeds Spearman on both model rows because each gets a few extreme genes right and is otherwise flat. In RANK the two models are slightly ANTI-correlated.

**Fig 2 – why the accuracy comparison is degenerate**

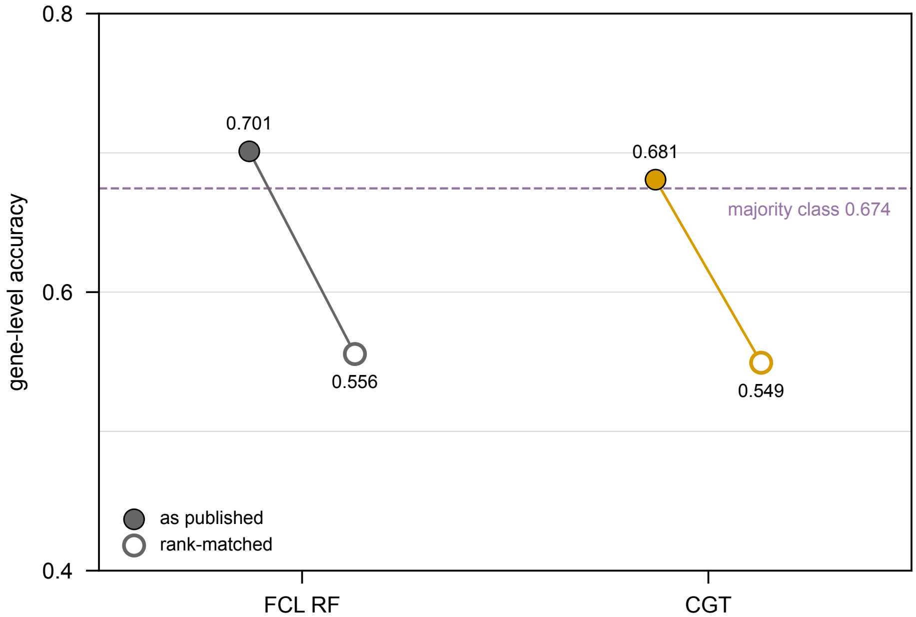


**Fig 2. Accuracy above the majority line is the conservative-majority strategy, not discrimination – for both models.** RF's 0.701 clears the 0.674 majority rate by 0.027 by calling 95 % of genes medium; CGT's absolute binning sits at 0.681 doing the same. Forced to the true class marginal, RF drops to 0.556 and CGT to 0.549. Evidence, not a result.

**Fig 3 – precision@k for high producers**

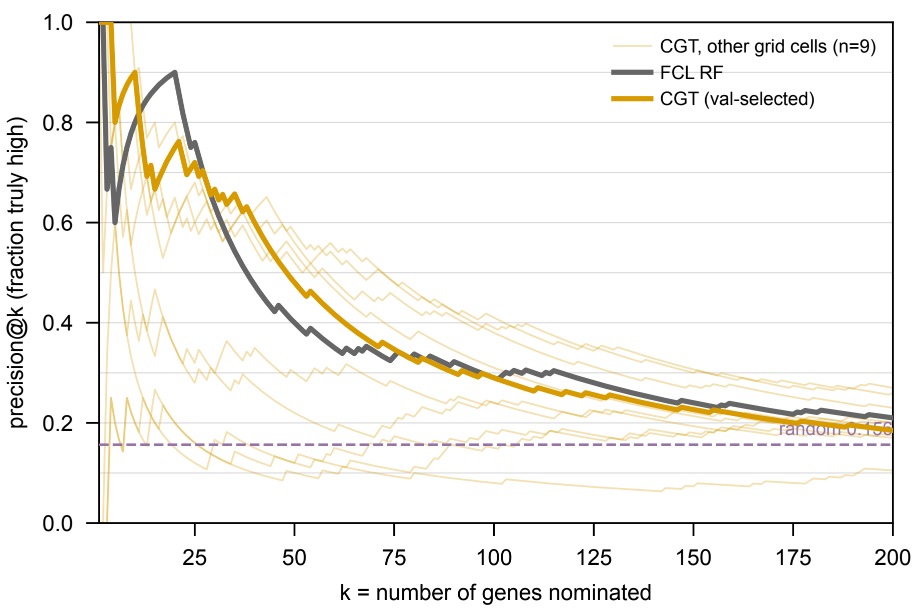


**Fig 3. Both models are strongly enriched at the top of their ranking.** Of the top k genes each nominates, the fraction truly high; random is 0.156.

| model | k=10 | k=25 | k=50 |
|---|---|---|---|
| FCL RF | 8 (5.1x, p=1e-05) | 19 (4.9x, p=8e-12) | 20 (2.6x, p=1e-05) |

| model | k=10 | k=25 | k=50 |
|---|---|---|---|
| CGT | **9** (5.8x, p=4e-07) | 18 (4.6x, p=1e-10) | **24** (3.1x, p=2e-08) |

Nine of CGT's top ten are genuine high producers. The two models track each other closely and both approach the base rate by k=150. The nine unselected CGT cells span 0.10-0.65 at k=50 – cell choice matters more than the CGT-vs-RF gap.

**Fig 4 – score by class, as percentile rank**

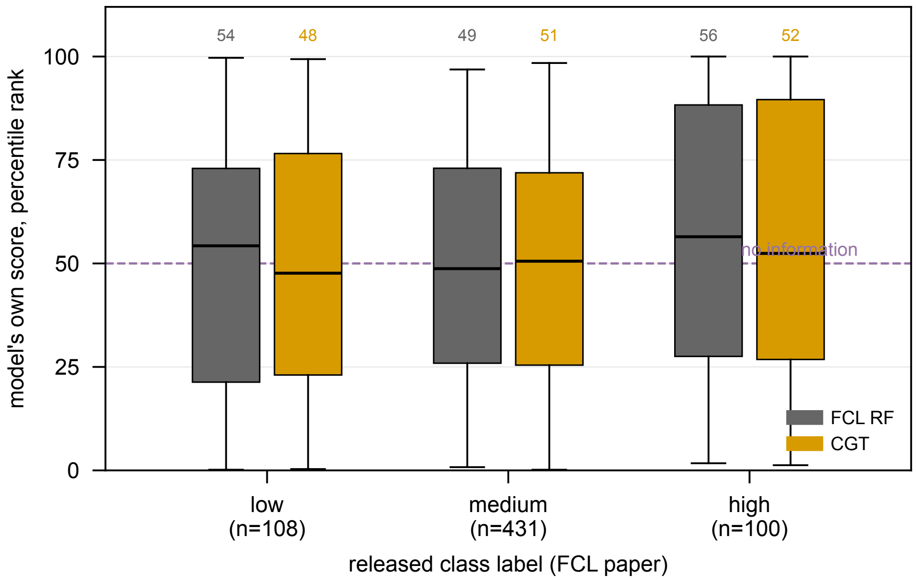


**Fig 4. Neither model orders the classes across the bulk of the population.** Percentile rank puts two models with incomparable units on one axis; a separating model shows three stepping boxes.

| model | low | medium | high |
|---|---|---|---|
| FCL RF | 54 | 49 | 56 |
| CGT | 48 | 51 | 52 |

**Read this WITH Fig 3, not instead of it.** A median is a bulk statistic and the top 25 genes are 4 % of the population, so it barely moves. Together the two figures say the signal is real and confined to the tail – not that there is none.

**Fig 5 – ROC for high-producer detection**

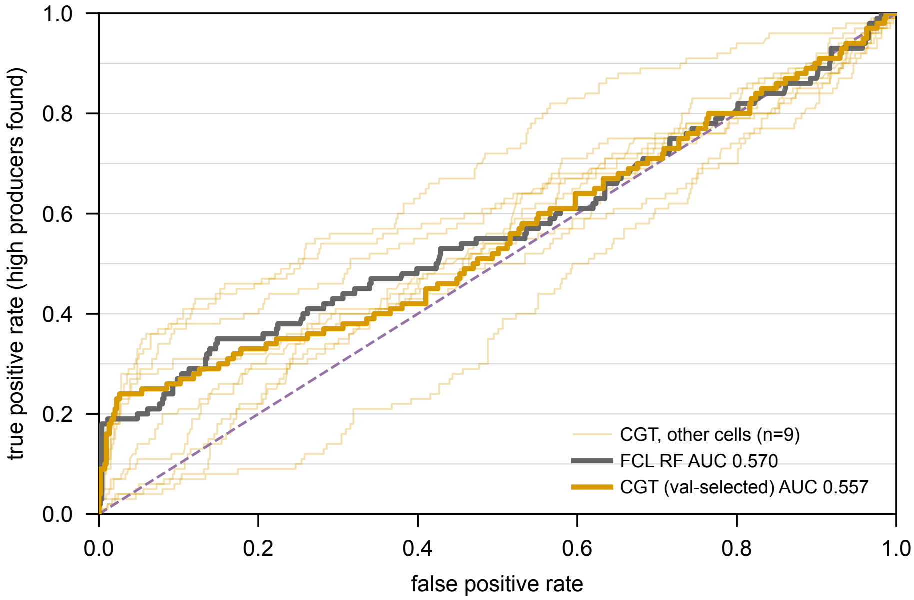


**Fig 5. Threshold-free, and on the val-selected cell it goes marginally to FCL: AUC 0.570 vs CGT's 0.557.** Both near the 0.500 chance line, because AUC integrates over the whole ranking where neither model has much.

**But four of the ten CGT cells DO beat FCL, and the best beats it clearly:**

| CGT cell | AUC | val pearson |
|---|---|---|
| `s08_L6_maskon_lr0.0001_zs` | **0.695** | 0.3528 |
| `s05_L2_maskoff_lr0.0001_yj` | 0.620 | 0.3326 |
| `s01_L2_maskon_lr0.0001_yj` | 0.618 | 0.2989 |
| `s04_L2_maskoff_lr0.0001_zs` | 0.571 | 0.3345 |
| **FCL RF** | **0.570** | – |
| `s09` (val-selected) | 0.557 | **0.3639** |

Note what this exposes: `s08` is the RUNNER-UP on validation (0.3528 vs `s09`'s 0.3639 – a 0.011 gap, well inside noise) and has an AUC 0.14 higher. **Validation pearson does not track test AUC**, so the val-selected cell landing below FCL here is close to a coin flip rather than a finding. 0.695 is the max over ten cells and cannot be quoted as "CGT's AUC" – but "CGT's AUC is 0.557, below FCL" is equally unsafe in the other direction. The honest statement is that this metric does not separate the two methods at the resolution we have.

**Fig 6 – how much is cell selection?**

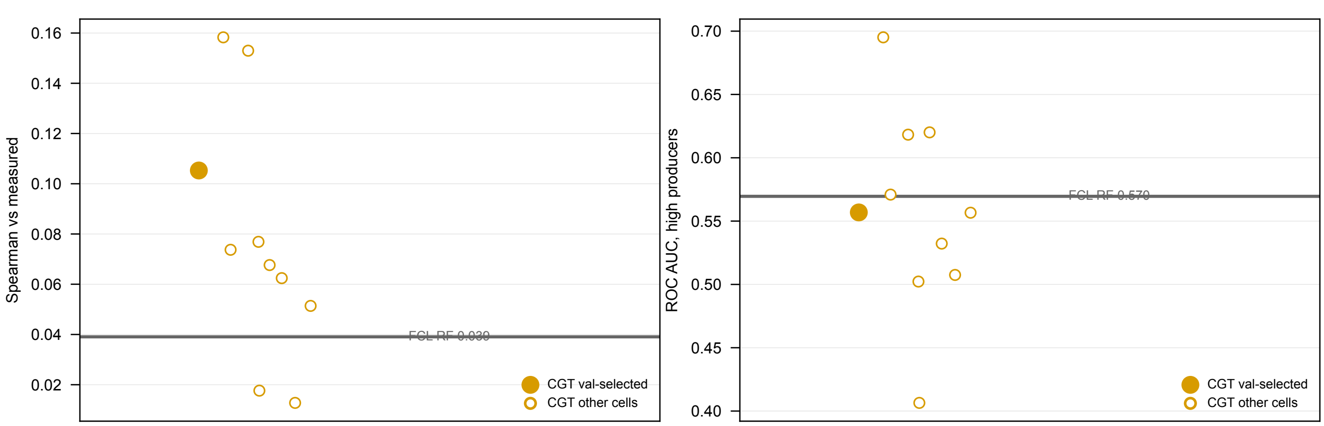

**Fig 6. The spread across grid cells is wider than the gap between models.** CGT cells span 0.013-0.158 Spearman (8 of 10 beat RF's 0.039) and 0.406-0.695 AUC (only 3 of 10 beat RF's 0.570, and the val-selected cell is not among them). Four low points are the lr=1e-3 cells that collapsed to a constant predictor. Any single-number claim is dominated by which cell is quoted, which is why the cell is fixed by validation in advance.

**Fig 7 – do their labels track our copy of the screen?**

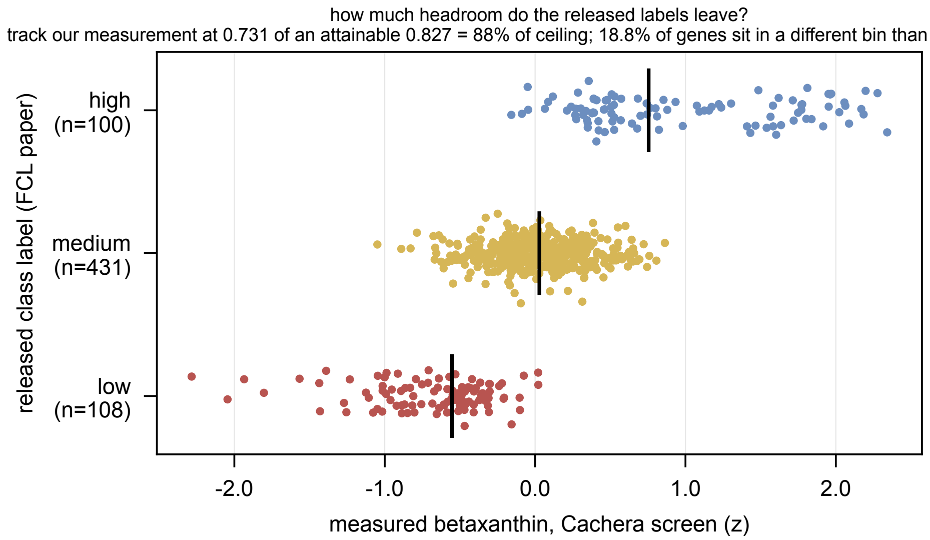


**Fig 7. This is a HEADROOM bound on the scoring, not a claim that the FCL paper binned anything wrongly.** Their labels are a correct application of their own rule; the rule min-max scales production, and their (smaller) copy of the screen has different extremes than ours, so the same rule lands its cuts in slightly different places. Neither labelling is "right" – theirs is the released ground truth and we score against it. The question the figure answers is how much of the residual error any model is charged with is really label disagreement. Spearman 0.731 against a ceiling of 0.827 (a 3-level ordinal cannot reach 1.0 against a continuous variable because of its own ties, so the raw number is uninterpretable alone) – **88 % of attainable, i.e. the labels track our measurement well.** The residual: **18.8 % of genes sit in a different bin than our values would give**, so a model that predicted OUR measurement perfectly would still be marked wrong on ~19 % of genes. That is the ceiling every number in this note sits under.

**Fig 8 – how little of the screen a flux-based method can reach**

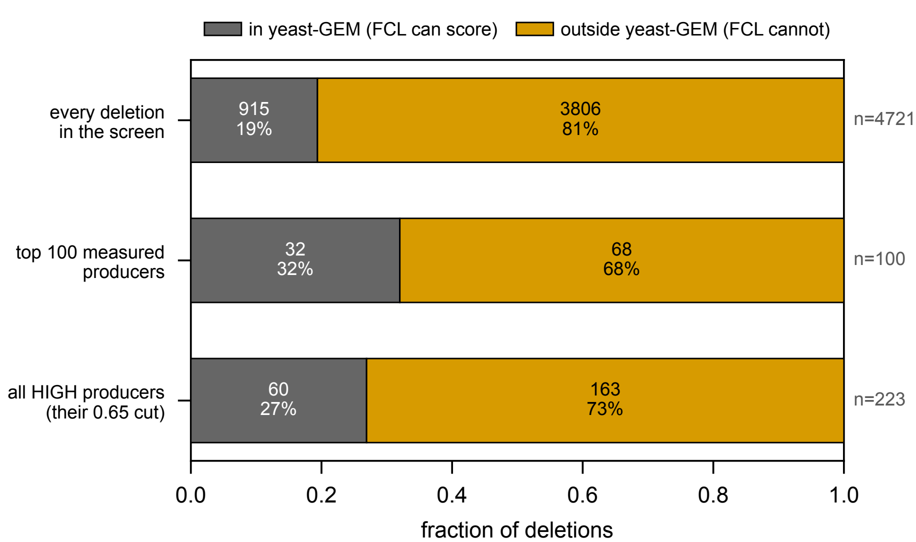


**Fig 8. yeast-GEM covers 19 % of the screen and misses 73 % of its high producers.** Measured data only – no model, no prediction. yeast-GEM contains 1,161 genes, of which **915 appear in the Cachera screen:**

**19.4 % of its 4,721 deletions**. A flux-based method has features for those and for nothing else.

| population | in yeast-GEM | outside |
|---|---|---|
| every deletion in the screen | 915 (19%) | **3,806 (81%)** |
| top 100 measured producers | 32 (32%) | **68 (68%)** |
| all HIGH producers (their 0.65 cut) | 60 (27%) | **163 (73%)** |

**The counter-point, stated honestly because it is real:** metabolic genes ARE enriched among high producers – 26.9 % of the highs against 19.4 % of the screen, a **1.4x enrichment**, exactly what the biology predicts. It is nowhere near enough to matter. They are such a small share of the genome that the large majority of high producers still sits outside the model. Restricting to metabolism buys a 1.4x concentration and costs three quarters of the hits.

This is what motivates a genome-scale model rather than a metabolism-only one, and Fig 9 asks whether CGT actually delivers on that reach.

### Fig 9 – the capability gap: metabolic vs non-metabolic genes

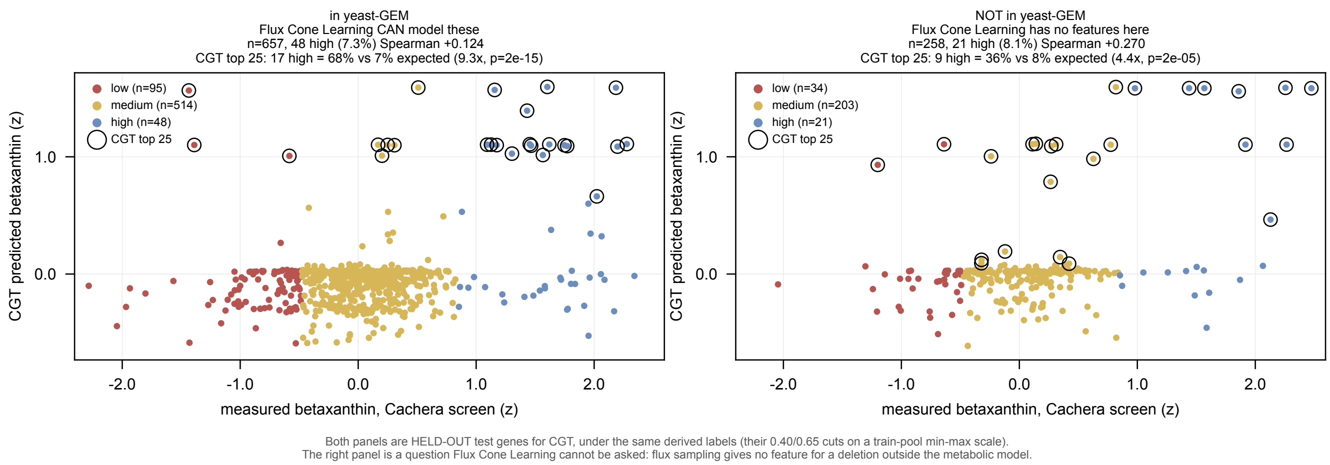


**Fig 8. CGT predicts high betaxanthin producers among deletions the flux-based method cannot represent.** Held-out test genes split by yeast-GEM membership, identical derived labels on both sides (their 0.40/0.65 cuts on a train-pool min-max scale). Circled = CGT's top 25.

| | in yeast-GEM | NOT in yeast-GEM |
|---|---|---|
| genes | 657 | 258 |
| true high (derived) | 48 (7.3%) | 21 (8.1%) |
| CGT Spearman vs measured | +0.124 | **+0.270** |
| CGT top 25: high found | 17 (68%) | 9 (36%) |
| enrichment over base rate | 9.3x, p=2e-15 | **4.4x, p=2e-05** |
| **Flux Cone Learning prediction** | available | **none possible** |

the FCL paper's test set is **639/640 (99.8 %) inside yeast-GEM**; of our other test genes only **21/294 (7.1 %)** are. That is what the constraint looks like in practice: flux sampling yields no feature for a deletion the metabolic model does not contain, so FCL's output for those genes is not poor, it is undefined.

CGT's rank correlation is *higher* on the non-metabolic set (+0.270 vs +0.124). Both enrichments are strongly significant.

**Caveats, and they matter for how hard this can be pushed.** (i) Labels for the non-GEM genes are DERIVED by us – the FCL paper never labelled them – so both panels use our derivation and the comparison to make is panel-vs-panel, not panel-vs-their-paper. (ii) n=258 with 21 positives is modest; the interval on 4.4x is wide. (iii) "FCL cannot model these" is a STRUCTURAL claim about flux-sample features, not an experiment we ran on their code.

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**Method notes**

**Can their side be ranked?** Yes – they release more than classes. `fig4c_*.csv` carries `score0/score1/score2` (P(low)/P(med)/P(high), summing to 1.0) for 124 flux samples per gene across 640 genes, aggregating to 558 distinct gene-level values via `E[class] = p_med + 2*p_high`. Two checks: TIES ARE INERT (82 genes share a value, but over 200 random permutations accuracy moves 0.5563 +- 0.0008 and high recall not at all); and THE AGGREGATION IS NOT LOAD-BEARING, the one used being the most generous to them – `E[class]` 0.5556/0.290, `p_high - p_low` identical, `p_high` alone 0.5446/0.280.

**The rank-matched numbers use an ORACLE class marginal.** Counts come from the TRUE TEST labels (108/431/100), so both models are handed the marginal of the answer. The sibling `evaluate_merzbacher_head_to_head.py` uses the TRAIN-POOL marginal (107/504/28) and is leak-free. Same marginal on both sides, so the RELATIVE claim holds; the ABSOLUTE values are not deployable accuracy:

| marginal | RF acc | CGT acc | RF hi-rec | CGT hi-rec |
|---|---|---|---|---|
| test counts (oracle, used here) | 0.5556 | 0.5493 | 0.290 | 0.290 |
| train-pool counts (leak-free) | 0.6228 | 0.6166 | 0.190 | 0.190 |

RF leads by ~0.006 either way, high recall identical either way. The pool expects 28 high producers where the test set holds 100: their test genes are enriched for extremes.

**Diagnostics not carried as figures** – regenerate with `--all`; they are current as of this render and sit alongside the others as `merzbacher_diag_*`: class distribution, per-class recall, rank-matched confusion matrices. Rank-matched, CGT reaches 0.29 recall on high producers vs their published 0.18 and 0.19 on low vs 0.06, paid for in medium recall (0.70 vs 0.98); the rank-binned labels agree on only 52 % of genes.

**Caveat on MCC.** The 0.205 quoted elsewhere is their PER-FLUX-SAMPLE MCC; they never published a gene-level MCC. Accuracy and high-producer recall are gene-level on both sides.

**Status.** 10 of 24 Delta `020_v4` cells finished; wave C (L=4) just started. Re-running the rsync and the scripts picks up the rest, and the CGT points can move.

**Slide set:** Fig 8 is the argument (coverage, not accuracy). Fig 2 + Fig 3 set it up – the published metric is empty, but both models do find high producers at the top. Fig 4 and Fig 6 are the qualifiers.